Standard LP (Linear Programming)

$$\min \ f = c^T x$$

$$\text{s.t.} \ \ Ax = b \geq 0, \quad x \geq 0$$

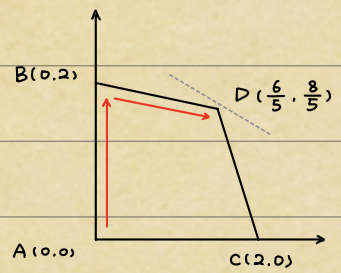
Inequality $\quad x_1 + x_2 \geq 2 \quad \Rightarrow \quad \begin{array}{l} x_1 + x_2 - S = 2 \\ S \geq 0 \end{array}$

没有 $x \geq 0$ $\quad x_1 \in \mathbb{R} \quad \Rightarrow \quad x_1^+ - x_1^- \geq 0$

Simplex

e.g. $\quad \max \ \ x_1 + 2x_2$

$$\begin{aligned} \text{s.t.} \ \ & 2x_1 + x_2 \leq 4 \\ & x_1 + 3x_2 \leq 6 \\ & x_1, x_2 \geq 0 \end{aligned}$$



先化标准型 $\quad \min \ \ -x_1 - 2x_2$

$$\begin{aligned} \text{s.t.} \ \ & 2x_1 + x_2 + S_1 = 4 \\ & x_1 + 3x_2 + S_2 = 6 \\ & x_1, x_2, S_1, S_2 \geq 0 \end{aligned}$$

$$\begin{bmatrix} 2 & 1 & 1 & 0 \\ 1 & 3 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ S_1 \\ S_2 \end{bmatrix} = \begin{bmatrix} 4 \\ 6 \end{bmatrix}$$

先让 2 个变量 x_1, x_2 为 0 $\quad$ Nonbasic Variable

再解出 S_1, S_2 $\quad$ Basic Variable

$(x_1 \ x_2 \ S_1 \ S_2)^{(0)} = (0, 0, 4, 6)$

由 $f = -x_1 - 2x_2$，变动 x_2 能让 f 减得更快

故 x_2 入基，即让 x_2 从 0 开始增加

最多到 $\min\{4/1, 6/3\} = 2$，此时 $S_2 = 0$，即 S_2 出基